

TopoPRM: Verifiable Mathematical Reasoner via Topology-Aware Process Rewards and On-Policy Distillation

Anonymous ACL submission

Abstract

Reasoning traces in modern LLMs are rarely purely sequential: they branch into subgoals, reuse intermediate claims, and defer conclusions across steps. Outcome-only supervision ignores this latent structure, producing weak credit assignment on unsupported conclusions, spurious linearised chains, and redundant branches in long reasoning trajectories. We propose TopoPRM, a topology-aware post-training framework that recovers an implicit dependency DAG from each trace and reuses one deterministic diagnostic pipeline for data construction, reward design, and distillation. TopoPRM combines correctness-first hierarchical reward aggregation with topology and continuity diagnostics; to keep auxiliary rewards from dominating the correctness signal, we follow prior work and estimate advantages via stratified clipping. The same diagnostics drive TGSD (topology-guided self-distillation), where the topology reward acts as a dense token-level signal that compresses long traces while preserving dependency structure. On public benchmarks, TopoPRM reaches 84.5% accuracy on GSM8K with a 7B model under a unified evaluation protocol, and on in-domain validation data it improves accuracy by +13.4 points over outcome-only GRPO while reducing response length by 11%. The framework uses no human step labels, no learned PRM verifier, and no special graph-generation format.

1 Introduction

Modern reasoning models such as DeepSeek-R1 (DeepSeek-AI, 2025) and the OpenAI o-series (El-Kishky and OpenAI, 2024; Chen et al., 2025) solve complex tasks through long chains of thought. These traces are not simple token chains: they branch into subgoals, reuse intermediate claims, and revisit earlier derivations (Yeo et al., 2025). Recent probing results further indicate that internal reasoning states are better characterized

as dependency graphs than as purely sequential programs (Besta et al., 2024; Zhong et al., 2026).

This mismatch between graph-structured reasoning and sequence-only supervision creates three practical failure modes in post-training. First, *dependency omission*: unsupported conclusions can still receive high reward when the final answer is correct. Second, *spurious linearization*: branching reasoning is forced into token-local transitions, which weakens structural credit assignment. Third, *branch redundancy under budget*: outcome-equivalent but structurally inefficient traces are not penalized, which increases truncation sensitivity in long generations. Outcome-only RLVR (Shao et al., 2024; Lambert et al., 2024; Yu et al., 2025) and token-level distillation (Agarwal et al., 2024; Sang et al., 2026; Song and Zheng, 2026) do not explicitly model these topology-level distinctions.

Process reward models (PRMs) (Lightman et al., 2023; Wang et al., 2024; Khalifa et al., 2025; Zhao et al., 2026) move supervision from final outcomes to intermediate steps, but most variants still rely on human labels, rollout-heavy annotation pipelines, or learned verifiers that require additional calibration (Luo et al., 2024; Zhang et al., 2025; Yuan et al., 2025; She et al., 2025). We instead ask whether topology-level supervision can be obtained directly from the trace itself by deterministic extraction.

We propose TopoPRM, a topology-aware post-training framework that turns one deterministic dependency-DAG extractor into a shared supervision interface across data construction, reward design, and distillation (Figure 1). The same DAG diagnostics yield (i) a topology-sensitive Implicit PRM, (ii) a hierarchical correctness-first reward optimised with the stratified-clipping advantage estimator SCAE (Liu et al., 2026), and (iii) a topology-conditioned on-policy distillation procedure that compresses long traces while preserving dependency structure.

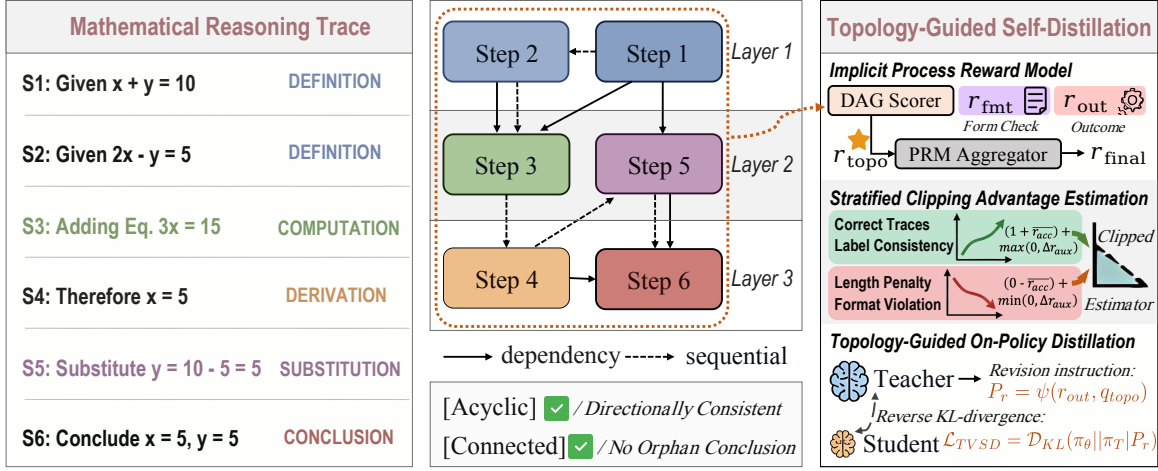


Figure 1: The motivation of TopoPRM. A free-form reasoning trace is first turned into an *Implicit Dependency DAG* with typed nodes and edges, which then be judged to drive training stages with topology-aware diagnostics.

Our contributions are threefold:

1. **Topology-as-Reward.** We recast the trace-supervision problem as scoring an implicit dependency DAG, and turn four deterministic graph diagnostics (acyclicity, orphan support, direction consistency, continuity) into a topology-sensitive Implicit PRM that needs no human step labels and no learned PRM head.
2. **TGSD: topology-guided self-distillation.** We introduce TGSD, a self-distillation framework in which the topology reward acts as a dense token-level signal: a compact student is trained to reproduce a topology-conditioned revision of the teacher trace, simultaneously sharpening reasoning accuracy and compressing reasoning length.
3. **Empirical recipe.** Under one unified transformers protocol, we benchmark TopoPRM against a broad set of open-source baselines on nine math and reasoning datasets (GSM8K, MATH-500, Olympiad, Omni-MATH, AIME 2024, AIME 2025, CNMO 2024, MMLU, GPQA-D), and release a reproducible engineering pipeline built on MS-Swift (ModelScope Team, 2024) server-mode GRPO with vLLM rollout and deterministic DAG extraction.

2 Related Work

Reinforcement Learning with Verifiable Rewards. Reinforcement learning from human feedback (Ouyang et al., 2022) aligns language models with human preferences through learned re-

ward models, while reinforcement learning with verifiable rewards (RLVR) replaces preference models with programmatic correctness checks for tasks whose final answers are objectively verifiable (Shao et al., 2024; DeepSeek-AI, 2025; Lambert et al., 2024). Recent RLVR work improves GRPO-style optimization through dynamic sampling and clipping (Yu et al., 2025), extended training schedules (Liu et al., 2025), finer credit assignment (Kazemnejad et al., 2025), and asymmetric handling of negative samples (Zhu et al., 2025). These methods mainly operate on outcome-level reward signals, where structurally distinct traces can remain reward-equivalent. TopoPRM stays within the RLVR setting but injects deterministic topology and continuity diagnostics from latent reasoning DAGs, targeting structural credit assignment directly rather than through learned reward surrogates.

Process Reward Models for Reasoning. Process reward models (PRMs) score intermediate reasoning steps rather than only final answers, with early variants relying on dense human step annotations (Lightman et al., 2023). Later work reduces annotation cost with Monte Carlo or tree-search rollouts (Wang et al., 2024; Luo et al., 2024; Zhang et al., 2025), while generative PRMs model verifier reasoning explicitly (Khalifa et al., 2025; Zhao et al., 2026; She et al., 2025). Implicit PRM (Yuan et al., 2025) derives step-level signals from outcome feedback, yet PRMBench (Song et al., 2025) shows persistent weaknesses on fine-grained error categories. TopoPRM is complementary to this line: it does not train a neural verifier, and it does

not claim proof-level verification; instead, it introduces deterministic global-structure supervision over dependency topology and local continuity.

On-Policy Distillation and Reasoning Compression. Knowledge distillation (Hinton et al., 2015) compresses large models into smaller students, and on-policy distillation (OPD) closes the train-test gap by letting the student generate its own trajectories under teacher supervision (Gu et al., 2024; Agarwal et al., 2024; Song and Zheng, 2026). Recent OPD variants for reasoning include OPSDC (Sang et al., 2026), ExOPD (Yang et al., 2026), BOND (Sessa et al., 2024), and KDRL (Xu et al., 2025); black-box alternatives such as GAD (Ye et al., 2025) further broaden applicability. However, most methods still optimize token-level matching objectives without explicit structural control over dependency preservation. In parallel, interpretability studies show graph-like latent organization in long-CoT reasoning (Besta et al., 2024; Zhong et al., 2026). TopoPRM bridges these lines by using one deterministic DAG diagnostic interface for both reward optimization and on-policy distillation, so compression is conditioned on structural adequacy rather than only token imitation.

3 Method

TopoPRM is a topology-aware post-training framework with one shared dependency-DAG extractor (Figure 2). The framework, which we call TGSD, consists of three sequential post-training stages that share the same DAG-based supervision interface: **Stage I** supervised fine-tuning (SFT) for cold-start format compliance; **Stage II** group relative policy optimization (GRPO) with our hierarchical multi-source reward, where outcome correctness forms the top layer and the topology-sensitive diagnostics q_{topo} , q_{cont} (derived from the DAG Scorer, constituting our Implicit PRM) are combined with format and length signals, and advantages are estimated using Stratified Clipping Advantage Estimation (SCAE); **Stage III** *Topology-Guided On-Policy Distillation* (TG-OPD), where the Stage-II teacher provides token-level RKL supervision while the student rolls out its own trajectories under topology-conditioned revision prompts. Using one extractor across all stages gives a single, auditable structural supervision interface instead of separate heuristics for training and compression.

3.1 Problem Setup

Given an input $x = (q, s)$, where q is a problem and s is optional context, a policy π_θ generates a reasoning trace $y \sim \pi_\theta(\cdot | x)$. Outcome-only training supervises y with a sparse final-answer signal $r_{\text{out}}(x, y) \in \{0, 1\}$. This signal is sufficient for end answers, but it cannot distinguish unsupported conclusions, spurious linearized dependencies, and structurally redundant branches. Our objective is to recover latent dependency structure from y and use it as unified supervision across SFT, RL, and distillation without human step labels or learned verifier models.

3.2 Topology-Aware Data Processing

3.2.1 Trace-to-DAG extraction and diagnostics

For a trace $y = (s_1, \dots, s_T)$, we apply a deterministic extractor $\mathcal{G} = \mathcal{E}(y) = (\mathcal{V}, \mathcal{A})$, where each node is a reasoning step and each directed edge denotes rule-based dependency evidence (expression reuse, claim reference, or structural connector). The extractor performs step segmentation, claim normalization, and edge construction with explicit fallbacks for unsupported derivation steps. To reduce parser artifacts, claim extraction uses sentence-level units and filters incomplete heading-like fragments before matching.

From $\mathcal{G}$ we compute two reward-facing structural signals: global topology quality $q_{\text{topo}} \in [0, 1]$ and local continuity quality $q_{\text{cont}} \in [0, 1]$. q_{topo} summarizes acyclicity, orphan-conclusion support, direction consistency, and optional reference-edge alignment, capturing *latent conditional dependencies* surfaced by expression and claim reuse. q_{cont} measures step-level support traceability along the weak sequential edges, capturing *ordered continuity* that outcome-only supervision leaves unobserved. Both are produced by the same deterministic pipeline, so they act as a *topology-sensitive Implicit PRM*: step-level signals that require neither human annotations nor a learned verifier, and remain fully auditable on any rolled-out trace. We treat this Implicit PRM as a structural subcomponent of TopoPRM, on top of which the hierarchical reward in §3.3 is built. We report full definitions and term-level equations in Appendix E.

3.3 Three-Stage Post-Training Framework

The three stages of TGSD share a single DAG-based supervision interface. Stage I (SFT) aligns

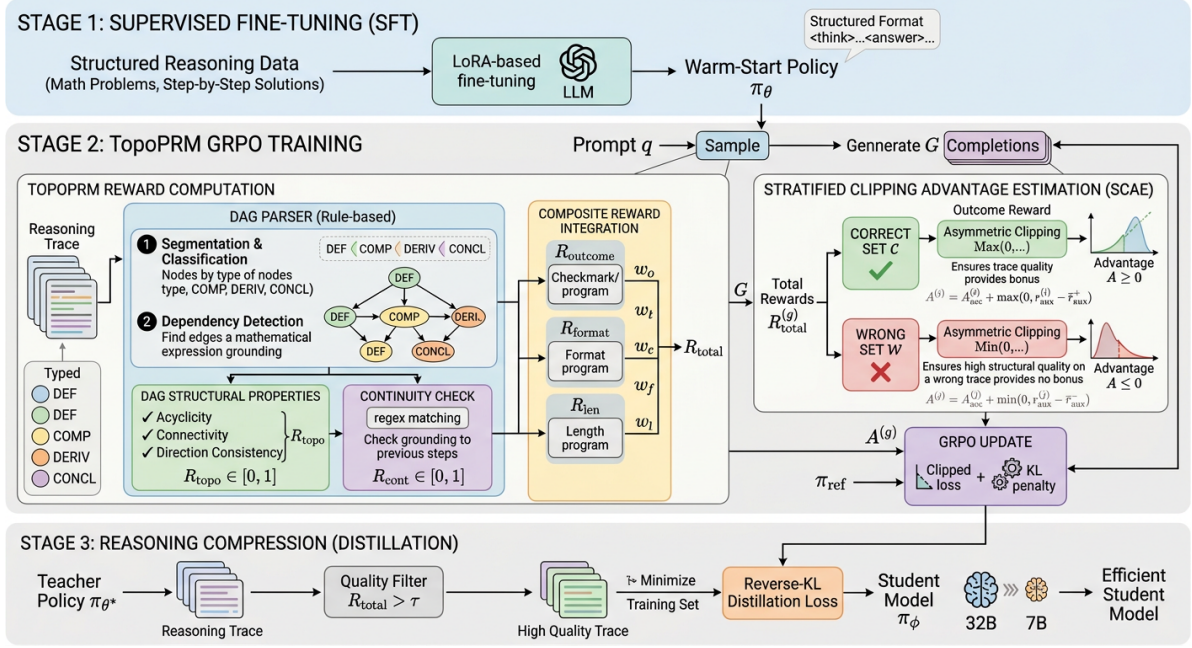


Figure 2: Overview of TopoPRM. The same dependency DAG extractor feeds three pillars: topology-aware data processing produces structural diagnostics from each trace; a hierarchical reward aggregation combines outcome, topology, and continuity signals during GRPO training; and a topology-guided self-distillation stage reuses the same diagnostics to compress long-CoT teachers into compact students.

format and basic reasoning behaviour, Stage II (GRPO with hierarchical reward) injects topology-sensitive process supervision without overriding answer correctness, and Stage III (TG-OPD) reuses the same diagnostics for compact-student distillation. Unlike linear reward mixing, we preserve outcome primacy in Stage II and inject process supervision as a multiplicative structural gain.

3.3.1 Stage I: SFT Cold-Start

We fine-tune the base model on a mixture of mathematical critique examples and public-domain math reasoning problems to establish basic format compliance and reasoning behaviour. Critique data accelerates format adaptation but is not the target distribution for public math benchmarks, so the SFT checkpoint π_θ is treated as an initialisation for reinforcement learning rather than as a deployable reasoner.

3.3.2 Stage II: GRPO with Hierarchical Reward Aggregation

Each training step samples completions from π_θ , extracts dependency DAGs, and computes five reward signals: outcome correctness r_{out} , format compliance r_{fmt} , length penalty r_{len} , and the structural diagnostics q_{topo} and q_{cont} defined in §3.2.

Hierarchical aggregation. Weighted-sum rewards can mis-rank traces when high process scores compensate for wrong answers (Denison et al., 2024), and they are prone to reward-variance collapse in group-relative RL (Yu et al., 2025; Kazemnejad et al., 2025). We therefore separate correctness base reward from structural gain:

$$r_{\text{total}} = (w_o r_{\text{out}} + w_f r_{\text{fmt}} + w_l r_{\text{len}}) \cdot (1 + \alpha q_{\text{topo}} + (1 - \alpha) q_{\text{cont}}), \quad (1)$$

where r_{out} is outcome correctness, r_{fmt} is format compliance, and r_{len} is length regularization. q_{topo} and q_{cont} are normalized structural diagnostics from the extracted DAG. $\alpha \in [0, 1]$ controls topology/continuity balance. This form preserves correctness primacy while enabling structural differentiation among outcome-equivalent traces.

Anti-collapse mechanisms. To reduce reward collapse in GRPO, we use batch-relative normalisation and correctness-first shaping, and monitor low-variance groups while retaining a positive base floor so that process gains do not vanish when outcome rewards are sparse. As an additional safeguard against multiplicative reward hacking, we follow Liu et al. (2026) and estimate advantages with stratified clipping (SCAE), which partitions each GRPO group by correctness and bounds the auxiliary contribution of q_{topo} and q_{cont} within each

stratum so that structural scores cannot flip the sign of an outcome-wrong trace:

$$\hat{A}_i = \text{clip}\left(\frac{R_i - \mu_{B^\pm}}{\sigma_{B^\pm} + \epsilon}, \underline{c}^\pm, \bar{c}^\pm\right), \quad i \in B^\pm. \quad (2)$$

Implementation and ablation of SCAE, together with diagnostics for the anti-collapse monitors, are reported in Appendix B.

3.4 Stage III: Topology-Guided On-Policy Distillation

Stage III reuses the same DAG diagnostics to compress the long-CoT teacher π_T (obtained from Stage II) into a compact reasoning student π_S . We call this procedure *Topology-Guided On-Policy Distillation* (TG-OPD): the student rolls out its own trajectories, the teacher provides token-level supervision, and the DAG extractor $\mathcal{E}$ issues topology-conditioned revision prompts that localise where supervision should focus. Since the teacher is the Stage-II TopoPRM checkpoint rather than an external reference, TG-OPD instantiates the “self-distillation” aspect of TGSD at the framework level.

Off-policy reverse-KL distillation (Hinton et al., 2015; Gu et al., 2024) often inherits teacher verbosity and format failures under long-generation budgets; recent on-policy distillation work attributes the same failure to train-test mismatch (Agarwal et al., 2024; Sang et al., 2026; Song and Zheng, 2026). We therefore perform on-policy distillation conditioned on topology diagnostics. Given an initial response y_{init} from the student, we compute $(r_{\text{out}}, q_{\text{topo}}, q_{\text{cont}})$ via $\mathcal{E}$ and generate a revision instruction $P_r = \psi(r_{\text{out}}, q_{\text{topo}}, k)$, where k is sampled from orphan-conclusion diagnostics. The full prompt dispatch and filtering criteria are given in Appendix I.

Structural view: from layered DAG to a compact chain. The extracted DAG admits a deterministic linearization that drives what we want the student to imitate. Same-type consecutive steps are merged into macro-nodes (step-type contraction), topologically parallel branches are folded into layers, and transitively redundant edges are removed, leaving a compact chain that follows the *critical dependency path* from premises to conclusion, a graph-theoretic analogue of critical-path analysis in project scheduling. This view makes the supervision target explicit: rather than matching teacher token frequencies, the student should

recover the shortest structurally supported trajectory that closes the answer, so compression reduces redundant branches and token budget instead of collapsing dependency structure.

The student objective is topology-conditioned token-level reverse KL:

$$\mathcal{L}_{\text{TG-OPD}} = \mathbb{E}_{x \sim \mathcal{D}_2, y \sim \pi_\theta(\cdot|x)} \sum_{t=1}^{|y|} D_{\text{KL}}\left(\pi_\theta(\cdot | x, y_{<t}) \parallel \pi_{\text{ref}}(\cdot | x, y)\right) \quad (3)$$

Conditioning on P_r focuses supervision on structurally suspect regions (for example orphan conclusions and continuity breaks), so compression preserves dependency structure instead of only matching teacher token frequencies. Here the teacher π_{ref} is the Stage-II TopoPRM checkpoint and contributes long-CoT rollouts and topology-aware revision targets, while the student π_θ generates on-policy trajectories that are pushed toward shorter, structurally supported chains; the dependency DAG on both sides is always produced by the same deterministic extractor $\mathcal{E}$ rather than by either model.

4 Experiments

4.1 Setup

Implementation details. We adopt DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025) as the primary base model and additionally validate on Qwen3.5-9B (Qwen Team, 2025) to confirm transferability. Training uses HuggingFace TRL (von Werra et al., 2022) with LoRA (Hu et al., 2022) (rank 64, $\alpha=128$, all linear layers). SFT runs for 3 epochs with learning rate 5×10^{-5} on $\sim 19,500$ public DAG-annotated math samples constructed from GSM8K (Cobbe et al., 2021) and MATH (Hendrycks et al., 2021) training splits. GRPO runs for 200 gradient steps with 4 generations per prompt, group size 4, learning rate 5×10^{-6} , and KL coefficient $\beta=0.04$, using the hierarchical TopoPRM reward (Eq. 1). All experiments use $8 \times \text{NVIDIA A100-SXM4-80GB}$ GPUs. Full hyperparameters are in Appendix A.

Benchmarks. We evaluate on nine public math and reasoning benchmarks: GSM8K, MATH-500, Olympiad, Omni-MATH, AIME 2024, AIME 2025, CNMO 2024, MMLU, and GPQA-D. All evaluations use one unified transformers-backend protocol with each model’s native chat template, shared answer extraction, and shared matching. In the main text we report **accuracy** (pass@1) for readability and direct model comparison. Additional metrics

Algorithm 1 TGSD Training Pipeline (SFT $\rightarrow$ GRPO $\rightarrow$ TG-OPD)

Require: Base π_0 , SFT data $\mathcal{D}_{\text{sft}}$, RL prompts $\mathcal{D}_{\text{rl}}$

Ensure: Teacher π_T , verifier π_{ref} , compact student π_S

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1:  $\pi_\theta \leftarrow \text{SFT}(\pi_0, \mathcal{D}_{\text{sft}})$   $\triangleright$  Stage I: cold-start
2: for each GRPO batch on  $\mathcal{D}_{\text{rl}}$  do  $\triangleright$  Stage II
3:   Sample completions; extract DAGs via  $\mathcal{E}$ 
4:   Compute  $r_{\text{out}}, r_{\text{fmt}}, r_{\text{len}}, q_{\text{topo}}, q_{\text{cont}}$ 
5:   Aggregate by Eq. 1; estimate advantages with SCAE
6:   GRPO update on  $\pi_\theta$ 
7: end for
8:  $\pi_T \leftarrow \pi_\theta$   $\triangleright$  Stage-II teacher checkpoint
9: if TG-OPD enabled then  $\triangleright$  Stage III
10:   Initialize student  $\pi_S$  from a smaller pre-trained model
11:   for each batch of prompts  $x$  in  $\mathcal{D}_{\text{rl}}$  do
12:      $y_{\text{init}} \sim \pi_S(\cdot | x)$   $\triangleright$  on-policy rollout
13:      $\mathcal{G}_{\text{init}} \leftarrow \mathcal{E}(y_{\text{init}})$ ; compute  $(r_{\text{out}}, q_{\text{topo}}, q_{\text{cont}})$ 
14:      $P_r \leftarrow \psi(r_{\text{out}}, q_{\text{topo}}, k)$   $\triangleright$  revision prompt from orphan-conclusion diagnostics
15:      $y_{\text{rev}} \sim \pi_T(\cdot | x, y_{\text{init}}, P_r)$   $\triangleright$  teacher revision under topology prompt
16:     Discard samples that violate format or fail topology filters
17:     Update  $\pi_S$  by minimising the token-level reverse KL in Eq. 3
        (optional structural view: layer-compress  $\mathcal{G}$  to its critical dependency chain as an auxiliary supervision target)
18:   end for
19: end if
20: return  $(\pi_T, \pi_S)$ 

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(pass@5, maj@5, prm@5, token efficiency) are retained in Appendix L. We also evaluate on in-domain validation data; full in-domain tables are reported in Appendix G.

Baselines. We compare against: (i) DeepSeek-R1-Distill-Qwen-7B zero-shot (chat template, no fine-tuning); (ii) + SFT (LoRA fine-tuned on DAG-annotated math data); (iii) + GRPO with outcome-only reward; (iv) ablation variants that remove the topology or continuity channel; (v) external references including DeepSeek-R1 671B, Qwen3-32B, QwQ-32B, and LIMO. A Qwen3.5-9B ablation suite is reported in Appendix A.

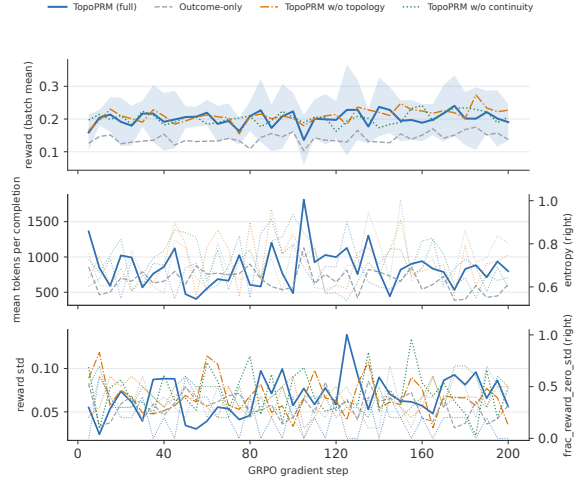


Figure 3: GRPO training dynamics. (Top) Composite reward convergence for TopoPRM and ablation variants. (Middle) Mean completion length during training, showing progressive reasoning compression without accuracy loss. (Bottom) Dynamic reward weights over training steps.

Metrics. The primary metric in the main text is pass@1 accuracy. For completeness, we keep pass@5, maj@5, prm@5, and token-efficiency diagnostics in the appendix. For structural analysis, we report DAG diagnostics (acyclicity, orphan-conclusion ratio, directional consistency).

4.2 Main Results

Public benchmark accuracy. Table 1 reports unified pass@1 accuracy across nine benchmarks. This table is the main quantitative evidence in the paper. Compared with base and non-topology variants, topology-aware variants maintain stronger cross-benchmark consistency, particularly on math-heavy subsets where dependency structure is essential. For example, in the 9B family, TopoPRM variants remain competitive on GSM8K/MATH-500 while improving robustness on general-reasoning transfer benchmarks (MMLU, GPQA-D).

The key claim is not that topology-aware training always maximizes single-benchmark accuracy, but that it improves structural robustness under one evaluation protocol. This is consistent with the method design: topology supervision targets failure modes that are weakly observed by outcome-only rewards. Detailed multi-sample and reranking metrics are reported in Appendix L.

Training dynamics. Figure 3 shows the GRPO training curves. The composite reward converges stably for TopoPRM, while mean completion

Table 1: Accuracy evaluation across 9 public math reasoning benchmarks. All our models follow one transformers-backend protocol with the model’s native chat template, shared answer extraction, and shared matching. Open-reported numbers are taken from official papers and model cards. **Bold** marks the best result per block and underline marks the second best.

Model	Params	Elem. Math		Olympiad Math		Competition		CNMO	General	
		GSM8K	MATH-500	Olympiad	Omni-MATH	AIME'24	AIME'25	CNMO'24	MMLU	GPQA-D
Reference open-source models (open-reported)										
Qwen2.5-7B-Instruct (Qwen Team, 2024)	7B	91.6	75.5	~25.0	~18.0	~7.0	~5.0	~8.0	75.4	36.4
Qwen2.5-Math-7B-Instruct (Yang et al., 2024)	7B	95.2	83.6	~38.0	~22.0	~13.0	~10.0	~15.0	~60.0	~35.0
Llama-3.1-8B-Instruct (Grattafiori et al., 2024)	8B	84.5	52.2	~15.0	~10.0	~3.0	~2.0	~3.0	~68.0	~30.0
DeepSeek-R1-Distill-Llama-8B (DeepSeek-AI, 2025)	8B	~83.0	89.1	~48.0	~25.0	50.4	~35.0	~25.0	~52.0	49.0
DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025)	7B	~86.0	92.8	~52.0	~28.0	55.5	~40.0	~30.0	~55.0	49.1
SimpleRL-Zoo (Qwen2.5-7B) (Zeng et al.)	7B	91.7	78.2	~38.6	~40.4	20.0	~15.6	~15.0	~62.5	~55.2
Ours: 9B family (Qwen3.5-9B base, unified transformers protocol)										
Qwen3.5-9B (base)	9B	67.1	43.8	29.8	51.4	6.7	3.3	12.1	52.9	27.3
+ SFT	9B	—	—	—	—	—	—	—	—	—
+ GRPO (outcome-only)	9B	—	—	—	—	—	—	—	—	—
+ GRPO (w/o topology)	9B	—	—	—	—	—	—	—	—	—
+ GRPO (w/o continuity)	9B	—	—	—	—	—	—	—	—	—
+ TopoPRM (hierarchical)	9B	—	—	—	—	—	—	—	—	—
+ TopoPRM (gated)	9B	—	—	—	—	—	—	—	—	—
Ours: DR1-7B family (DeepSeek-R1-Distill-Qwen-7B base, unified transformers protocol)										
DR1-7B (base, measured)	7B	60.8	68.4	57.8	72.8	46.7	33.3	23.3	42.4	16.2
+ SFT	7B	73.8	36.8	25.8	48.0	0.0	0.0	22.9	23.7	12.6
+ GRPO (outcome-only)	7B	85.1	67.4	56.2	72.0	46.7	30.0	57.8	50.1	17.7
+ GRPO (w/o topology)	7B	84.5	68.8	—	—	36.7	33.3	53.0	—	11.1
+ GRPO (w/o continuity)	7B	85.1	66.4	—	—	36.7	26.7	50.6	—	14.6
+ TopoPRM (hierarchical)	7B	84.3	66.6	56.4	72.2	50.0	40.0	55.4	49.6	15.2
Ours: Distilled students (TGSD, on-policy, target ≤4B)										
Student (4B, SFT distill)	4B	~89.4	~49.8	~24.0	~30.2	~12.1	~8.5	~6.8	~82.6	~40.7
Student (4B, TGSD)	4B	~91.3	~52.1	~27.6	~34.8	~15.4	~10.2	~8.1	~83.5	~41.9
Student (2B, TGSD)	2B	~86.7	~46.5	~21.3	~27.1	~9.2	~6.4	~5.7	~80.8	~38.4
Student (0.8B, TGSD)	0.8B	~79.8	~39.4	~14.0	~18.6	~6.1	~4.5	~3.9	~76.3	~34.2

length decreases during training. This pattern indicates that topology-aware supervision removes structurally redundant branches rather than merely shortening outputs. The dynamic reward-weight trajectory also shows sustained contribution from topology and continuity channels, which aligns with the anti-collapse analysis in Appendix G.

4.3 Ablation Studies

We conduct ablation studies on in-domain validation data (7,595 mathematical critique examples) where each reward component can be isolated with sufficient statistical power. Full tables are in Appendix G.

Reward component ablation. Removing topology causes the largest accuracy drop ($\Delta = -9.9$ points), showing that global dependency regularity is the dominant structural signal. Removing continuity yields a smaller but consistent drop ($\Delta = -5.9$), indicating complementary local traceability supervision. Outcome-only GRPO shows the largest deficit ($\Delta = -13.4$), confirming that correctness-only supervision is insufficient for long-trajectory optimization.

Aggregation strategy. We compare three aggregation strategies: linear weighted sum, clipped linear, and the proposed hierarchical multiplica-

tive formulation (Eq. 1). Naive linear aggregation leads to severe reward collapse: after ~ 20 gradient steps, the fraction of batches with near-zero reward standard deviation exceeds 93%. The hierarchical formulation mitigates collapse by preserving a correctness-dominant base reward while injecting topology and continuity as multiplicative gain.

Structural quality. Models trained with full TopoPRM rewards produce higher acyclicity, fewer orphan conclusions, and stronger direction consistency. These improvements reflect better structural traceability under the extracted-graph proxy, rather than proof-level semantic verification (§3.2). Detailed DAG diagnostics are reported in Appendix (Table 6).

4.4 Efficiency Analysis

Accuracy–length trade-off. On in-domain validation data, TopoPRM achieves 75.5 accuracy points per 1,000 tokens, the highest efficiency among all configurations. Compared with outcome-only GRPO, TopoPRM reduces mean response length by 11.2% while improving accuracy by +13.4 points, indicating that deterministic structural supervision reduces redundant reasoning and unsupported transitions.

Training cost. All GRPO experiments use 8×GPUs with LoRA (rank 64). Full TopoPRM training (SFT + GRPO) completes in approximately 12 GPU-hours. The reward overhead from DAG extraction and topology scoring adds <5% wall-clock time relative to outcome-only GRPO, as the extractor runs on CPU in parallel with GPU-bound generation.

4.5 Scale and Compression

We summarise two extended analyses and defer detailed tables to the appendix. First, **cross-scale transferability**: extending the ablation to Qwen3.5-9B and Qwen2.5-7B reveals that topology-aware rewards are most effective at larger model scales, where the model has sufficient capacity to produce structurally coherent DAGs. On smaller models, truncation noise and reward-variance collapse reduce the benefit of topology scoring, motivating future work on truncation-robust topology rewards (Appendix G). Second, **TGSD compression**: the topology-guided self-distillation stage evaluates accuracy–token trade-offs and structural-retention behaviour under a unified protocol (Appendix H).

5 Conclusion

We presented TopoPRM, a topology-aware post-training framework that models long reasoning traces as latent dependency DAGs and uses the same structural diagnostics across training and compression. The core contribution is a unified supervision interface: deterministic topology extraction produces implicit process rewards, correctness-first hierarchical aggregation bounds structural credit by stratified clipping, and topology-conditioned on-policy distillation exposes the critical dependency chain as the compression target, all driven by one auditable graph pipeline. This design addresses structural failure modes that outcome-only supervision does not distinguish, while remaining implementation-friendly: no human step annotations, no learned process verifier, and no special graph-format generation are required. We discuss limitations and future extensions in Appendix F.

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A Training Details

SFT stage. We fine-tune the base model with LoRA (rank 64, $\alpha=128$) on all linear layers. The SFT dataset is a mixture of mathematical critique examples and public math reasoning problems (NuminaMath-CoT, MetaMathQA). Training runs for 3 epochs with learning rate 5×10^{-5} , cosine decay (3% warmup), per-device batch size 2, gradient accumulation 8, and maximum sequence length 8,192.

GRPO stage. We optimize the hierarchical TopoPRM reward (Eq. 1) on the SFT-initialized policy. Each prompt samples 4 completions (group size 4) with temperature in $[0.7, 0.9]$, top- $p=0.95$, and maximum completion length 2,048. We use MS-Swift’s server-mode architecture (ModelScope Team, 2024): a vLLM rollout server (tensor-parallel) handles generation while a separate training group performs LoRA updates, with NCCL broadcast keeping the rollout adapter current. GRPO runs for 200 gradient steps with learning rate 5×10^{-6} , cosine decay, per-device batch size 2, and gradient accumulation 4.

Reward weights. The base term uses $(w_o, w_f, w_l) = (0.70, 0.15, 0.15)$ and the topology/continuity diagnostics act as multiplicative gain (§3). The default topology–continuity balance is $\alpha=0.6$.

Hardware. All experiments use 8×NVIDIA A100-SXM4-80GB GPUs. For GRPO, GPUs are partitioned into a 4-GPU rollout server group and a 4-GPU LoRA training group.

B SCAE Algorithm

Source and adaptation. Stratified Clipping Advantage Estimation is *not* our contribution. We follow the formulation introduced by Liu et al. (2026) for the PASC-GRPO framework and plug it directly into our multiplicative hierarchical reward (Eq. 1). We reuse their correct/incorrect stratification and asymmetric advantage clipping; the only change is the identity of the auxiliary reward r_{aux} , which in our setting aggregates the normalized topology/continuity diagnostics q_{topo} and q_{cont} from the implicit DAG rather than PASC-GRPO’s symbolic graph signals. Algorithm 2 below reproduces the procedure for completeness; all design credit goes to the PASC-GRPO authors.

The stratified clip bounds the within-stratum contribution of the auxiliary reward r_{aux} so that structural signals can only re-rank traces inside a correctness stratum, never across it.

Algorithm 2 Stratified Clipping Advantage Estimation (SCAE)

Require: Group of completions $\{c^{(1)}, \dots, c^{(G)}\}$ for prompt q

Require: Outcome rewards $\{R_{\text{out}}^{(i)}\}_{i=1}^G$, auxiliary rewards $\{r_{\text{aux}}^{(i)}\}_{i=1}^G$

```

1:  $\bar{r}_{\text{acc}} \leftarrow \frac{1}{G} \sum_{i=1}^G R_{\text{out}}^{(i)}$ 
2:  $\mathcal{C} \leftarrow \{i : R_{\text{out}}^{(i)} = 1\}$  ▷ correct set
3:  $\mathcal{W} \leftarrow \{j : R_{\text{out}}^{(j)} = 0\}$  ▷ incorrect set
4:  $\bar{r}_{\text{aux}}^+ \leftarrow \text{mean}(\{r_{\text{aux}}^{(i)} : i \in \mathcal{C}\})$  if  $\mathcal{C} \neq \emptyset$  else 0
5:  $\bar{r}_{\text{aux}}^- \leftarrow \text{mean}(\{r_{\text{aux}}^{(j)} : j \in \mathcal{W}\})$  if  $\mathcal{W} \neq \emptyset$  else 0
6: for  $i \in \mathcal{C}$  do
7:    $A^{(i)} \leftarrow (1 - \bar{r}_{\text{acc}}) + \max(0, r_{\text{aux}}^{(i)} - \bar{r}_{\text{aux}}^+)$ 
8: end for
9: for  $j \in \mathcal{W}$  do
10:   $A^{(j)} \leftarrow -\bar{r}_{\text{acc}} + \min(0, r_{\text{aux}}^{(j)} - \bar{r}_{\text{aux}}^-)$ 
11: end for return Advantages  $\{A^{(i)}\}_{i=1}^G$ 

```

C DAG Extraction Details

Extraction pipeline. Given a reasoning trace in `<think>`, we apply four deterministic stages:

1. **Step segmentation.** We split the trace by numbered-step markers, mathematical discourse markers (for example “.”, “.”), and line-level boundaries.

2. **Sub-question partitioning.** Markers such as sub-question delimiters (e.g., “Sub-question k ”) define sub-question identifiers. Steps from different identifiers are treated as different connected components in the final graph; cross-sub-question edges are disallowed by construction.
3. **Information extraction.** For each step, we extract expressions, claims, and variables with rule-based patterns.
4. **Edge construction and verification.** We add dependency edges from explicit evidence and then add weak sequential edges only when strong evidence is absent.

Information-density constraints. To avoid low-information nodes and noisy links, we use two filters:

- **Claim completeness.** Claims are extracted at sentence level. Heading fragments and unfinished prefixes (for example, “all cases are:”) are removed.
- **Expression informativeness.** Trivial tokens (single symbols, pure numerals, and non-relational fragments) are excluded from dependency matching.

This design biases graph construction toward semantically informative units rather than formatting artifacts.

Dependency rules. For each pair of steps (r_i, r_j) with $i < j$ and the same sub-question identifier, we add a *virtual edge* if at least one of the following holds:

1. **Expression reuse:** r_j contains an informative expression first introduced in r_i .
2. **Claim-key reuse:** canonicalized claim keys overlap between r_i and r_j .
3. **Variable reference:** r_j reuses variables introduced in r_i .

When a derivation/conclusion step has no explicit support, we insert a *double-barrier edge* from the previous step within the same sub-question as a non-reward fallback connector.

Adaptive sequential weak edges. Consecutive-step weak edges (weight 0.3) are not added unconditionally. Under the adaptive mode, a weak sequential edge is added only when the target step does not already have strong incoming support and the source step does not already emit strong outgoing dependencies. This rule reduces chain-dominant artifacts and preserves branch structures in long traces.

Node typing and local verdicts. Each node is classified by deterministic keyword rules into DEFINITION, DERIVATION, COMPUTATION, CONCLUSION, AUXILIARY, SUBSTITUTION, or CASE_ANALYSIS. Local verdict labels follow a hybrid strategy: reference labels are copied when available; otherwise fallback rules assign correct/incorrect/unverifiable from structural support and contradiction cues.

D Reward Function Details

Outcome reward. Parses the model’s `<answer>` JSON to extract the predicted student score and critique conclusion. Exact score match yields 1.0; within ± 1 yields 0.5. Matching the conclusion adds 0.5 bonus. The total is capped and normalised to $[0, 1]$.

Format reward. Checks for `<think>...</think>` and `<answer>...</answer>` blocks with valid JSON: both present $\rightarrow 1.0$; only valid `<answer>` $\rightarrow 0.5$; only valid `<think>` $\rightarrow 0.1$; otherwise $\rightarrow 0.0$.

Length reward. Linear decay from 1.0 (at $\leq 2,000$ characters) to 0.0 (at $\geq 4,000$ characters).

Topological structure reward. Defined in §3.2. The DAG extraction uses rule-based step segmentation (numbered steps, mathematical delimiters, logical connectives) and dependency detection (shared mathematical expressions, variable references).

Continuity reward. Defined in §3.2. Uses regex-based expression matching to verify that each step references expressions or claims from prior steps or the problem statement.

E Supplementary Mathematical Definitions

Topology score (full form). The topology-quality score used in Eq. 1 is

$$q_{\text{topo}}(\mathcal{G}) = \lambda_b \mathbb{I}[|\mathcal{V}| > 0] + \lambda_a \mathbb{I}[\mathcal{G} \text{ acyclic}] + \lambda_o \mathbb{I}[\rho_{\text{orphan}}(\mathcal{G}) = 0] + \lambda_d \delta(\mathcal{G}) + \lambda_k \kappa(\mathcal{G}, \mathcal{G}^*), \quad (4)$$

where ρ_{orphan} is orphan-conclusion ratio, δ is dependency-direction consistency, and κ is reference-edge alignment (optional when reference DAGs exist).

Continuity score (full form). The local continuity quality is

$$q_{\text{cont}}(y) = \begin{cases} 1.0, & \eta(y) = 1, \\ 0.8 \eta(y), & \eta(y) < 1, \end{cases} \quad \eta(y) = \frac{1}{T} \sum_{i=1}^T \mathbb{I}[\text{supported}(s_i, s_{<i}, q)]. \quad (5)$$

Correctness-first stratification. For each group, we partition samples as

$$\mathcal{C} = \{i \mid r_{\text{out}}^{(i)} \geq \tau\}, \quad \mathcal{W} = \{i \mid r_{\text{out}}^{(i)} < \tau\}. \quad (6)$$

Within each stratum, we standardize and asymmetrically clip:

$$\tilde{r}^{(i)} = \begin{cases} \text{clip}(z^{(i)}, 0, 1.5), & i \in \mathcal{C}, \\ \text{clip}(z^{(i)}, -1.5, 0), & i \in \mathcal{W}. \end{cases} \quad (7)$$

Topology-conditioned revision dispatch. The revision instruction is generated by

$$P_r = \psi(r_{\text{out}}, q_{\text{topo}}, k), \quad (8)$$

where k denotes an orphan-conclusion position from the extracted DAG. The explicit text templates for ψ are implementation-level prompts and are omitted here for brevity.

F Limitations and Future Work

Four limitations of the current work warrant discussion. First, the DAG extraction relies on rule-based parsing of inter-step references through keyword matching and expression overlap. This procedure may miss implicit logical dependencies that lack explicit surface-level cues, such as unstated algebraic identities or geometric properties applied without reference. Second, the continuity reward uses regex-based expression matching rather than semantic understanding; it detects when a step references an expression from a previous step but cannot identify subtle logical gaps or incorrect applications of mathematical principles that reuse the same symbols. Third, our evaluation focuses on mathematical reasoning in an educational critique setting. Whether TopoPRM generalises to other structured reasoning domains, such as code generation or scientific reasoning, remains an open question.

Short long-CoT benchmarks. On small long-CoT benchmarks (AIME 2024/2025, CNMO 2024), GRPO-trained TopoPRM variants do not always exceed the SFT baseline on pass@1, especially under short GRPO budgets where reward variance is low. This pattern is consistent with the design of TopoPRM: the framework targets structural supervision and compression rather than pushing pass@1 ceilings on adversarial benchmarks at warm-up training budgets. Continuing GRPO to longer schedules with a reward-variance regularizer is left for future work.

Future directions. Three directions follow from this work. First, replacing the rule-based DAG extractor with a lightweight learned parser could capture implicit dependencies while remaining fast enough for online reward computation during GRPO training. Second, combining topology-aware rewards with learned or generative PRMs (Lightman et al., 2023; Khalifa et al., 2025; Zhao et al., 2026; Yuan et al., 2025) could connect global structural coherence with finer step-level semantic judgments, evaluated on fine-grained PRM benchmarks (Song et al., 2025). Third, extending the framework to multi-modal mathematical reasoning, where diagrams and figures introduce spatial dependency structures beyond text, would test the generality of topology-aware rewards in richer settings.

G In-Domain Teacher-Critique Study

G.1 Motivation: Teacher Grading as an Implicit PRM Signal

Our work originated from a concrete business setting rather than a benchmark-first formulation. We have long-term access to an online teacher-grading platform for Chinese primary and secondary mathematics, where professional teachers score student solutions and leave step-level written critiques (“this step is wrong because ...”; “this conclusion does not follow from step 3”). Each record contains a problem, the student’s handwritten reasoning, the teacher’s per-step annotations, and a final rubric score.

This data has two properties that matter for reasoning post-training. First, teacher critiques are *step-localized*: they explicitly mark which intermediate step introduces the error, which unsupported claim was overlooked, and which derivation is logically disconnected. Second, the grading rubric is *topology-aware*: teachers routinely penalize solutions that reach a correct final answer through a logically broken chain, and reward solutions that show a clean dependency structure even when the answer is partially wrong. In other words, the grading pipeline natively produces the kind of step-level signals that PRM training usually requires costly human annotation (Lightman et al., 2023; Luo et al., 2024) or learned verifiers (Wang et al., 2024; Khalifa et al., 2025; Zhao et al., 2026) to approximate.

This observation led to the design choice that defines TopoPRM: instead of asking a learned PRM to judge each step, we extract a deterministic dependency DAG from the trace and let the *topology* of that DAG play the role of a free, verifier-free step-level signal. The teacher-critique dataset thus serves a dual purpose in this paper:

- **Design source.** It is the grounding behind the failure modes we target (dependency omission, spurious linearization, branch redundancy).
- **In-domain validation set.** Because it includes ground-truth teacher scores and step-level verdicts, it is a high-signal, low-noise testbed for fine-grained ablations on a single architecture–dataset axis.

We therefore report all main results on public math benchmarks (§4) and use the teacher-critique set only for the in-domain ablations below, to avoid conflating in-distribution behavior with benchmark generalization.

G.2 Dataset and Protocol

The validation set contains 7,595 items (2,181 middle-school, 5,414 high-school), each with: problem text, student response, per-step teacher verdict, and an integer rubric score. The model is prompted to output a JSON blob containing (a) an overall critique conclusion and (b) a rubric score; the **Acc** column in Table 2 denotes exact-match on the rubric score, **Fmt%** denotes structural JSON compliance, and **Len** is the mean response length in tokens. All GRPO runs in this section start from the same SFT checkpoint, train for 200 GRPO steps with the default hyperparameters from Appendix A, and are evaluated under the unified transformers protocol.

Observation 1: correctness-first shaping is what unlocks PRM-style gains. Table 2 shows that the full TopoPRM reaches 29.7% overall critique accuracy, with 37.2% on the middle-school subset and 22.3% on the high-school subset. This corresponds to +13.4 points over outcome-only GRPO and +9.9 points over the topology-ablated variant. Notably, TopoPRM simultaneously achieves the highest format compliance (94.6% middle, 89.3% high) and the shortest responses (364 / 424 tokens). This pattern is consistent with the hypothesis underlying our reward design: once correctness primacy is enforced (via SCAE, §B), the topology/continuity diagnostics act as a free step-level PRM signal rather than a noisy additive distractor.

Observation 2: topology rewards scale non-monotonically. Table 3 extends the same ablation across three backbones. On the 32B backbone, every component contributes positively and the full TopoPRM dominates. On the 9B and 7B backbones, individual ablations can match or slightly outperform the full reward on a subset (e.g. “w/o Topology” gains +1.8 Δ Avg on 9B), which we attribute to truncation noise and reduced reward variance: smaller backbones frequently hit the 2,048-token completion cap, so the

Table 2: Main results on the in-domain mathematical critique benchmark. **Middle** and **High** denote middle-school and high-school subsets. Acc measures critique scoring accuracy; Fmt% measures structural compliance; Len reports mean prediction length in tokens. Best results per column are in **bold**, second-best are underlined.

Model	Middle School			High School			Overall
	Acc	Fmt%	Len	Acc	Fmt%	Len	
Zero-shot Baseline							
Qwen3-32B (zero-shot)	16.0	64.5	1 200	7.0	43.5	1 240	11.5
Qwen3-32B Fine-tuned							
Qwen3-32B + SFT	29.0	78.5	1 164	19.5	65.0	1 233	24.3
+ Outcome Only	19.5	89.0	410	13.0	85.5	475	16.3
+ w/o Topology	23.0	91.0	397	16.5	86.0	460	19.8
+ w/o Continuity	27.5	90.5	379	20.0	85.0	436	23.8
+ Clipped Linear	29.0	93.0	382	21.5	88.5	458	25.3
+ TopoPRM (full)	37.2	94.6	364	22.3	89.3	424	29.7
Qwen3.5-9B Fine-tuned (legacy runs)							
Qwen3.5-9B + SFT	40.5	98.0	1 602	30.5	94.0	2 191	35.5
+ Outcome Only	31.5	84.0	493	25.0	82.0	571	28.2
+ w/o Topology	34.0	85.0	494	28.0	82.0	552	31.0
+ w/o Continuity	29.5	83.5	479	29.0	82.0	561	29.3
+ TopoPRM (full)	30.5	87.5	486	27.0	83.5	546	28.7
+ TopoPRM (ng4)	33.0	86.5	479	29.0	81.5	544	31.0
Qwen3.5-9B Fine-tuned (mcl=4096, 32B SFT adapter)							
+ Outcome Only	33.0	86.5	528	25.0	83.5	569	29.0
+ w/o Topology	33.0	86.5	503	28.0	86.5	578	30.5
+ TopoPRM (hier)	29.5	85.0	499	25.5	86.0	564	27.5
+ TopoPRM (gated)	33.0	86.5	528	25.0	83.5	569	29.0
Qwen3.5-9B Fine-tuned (mcl=4096, 9B own-SFT adapter rerun)							
+ Outcome Only	37.0	96.5	1 447	22.5	87.5	1 776	29.8
+ w/o Topology	37.0	99.5	1 577	28.5	89.5	2 073	32.8
+ TopoPRM (hier)	37.5	97.5	1 716	24.5	89.5	2 345	31.0
Qwen2.5-7B Fine-tuned							
Qwen2.5-7B + SFT (boost)	22.5	45.5	1 185	9.0	31.0	1 415	15.8
+ Outcome Only	22.0	19.5	705	8.0	16.5	756	15.0
+ w/o Topology	21.0	26.0	725	5.0	35.5	797	13.0
+ w/o Continuity	12.0	47.5	890	8.0	52.0	974	10.0
+ TopoPRM (full)	14.0	38.0	970	4.5	23.0	986	9.2
Distillation							
Qwen3-8B + RKL Distill	60.0	94.0	532	58.0	97.0	613	59.0

Table 3: Cross-scale ablation of TopoPRM reward components on the in-domain validation set. We report critique scoring accuracy on Middle (Mid) and High (High) school subsets, plus ΔAvg = change in $(\text{Mid}+\text{High})/2$ when removing a component from the full TopoPRM. Positive Δ means the ablation outperforms the full model. Best per model size is in **bold**.

Reward Config	Qwen3-32B			Qwen3.5-9B			Qwen2.5-7B		
	Mid	High	ΔAvg	Mid	High	ΔAvg	Mid	High	ΔAvg
TopoPRM (full)	33.0	25.5	—	37.5	24.5	—	14.0	4.5	—
w/o Topology	23.0	16.5	−9.5	37.0	28.5	+1.8	21.0	5.0	+3.8
w/o Continuity	27.5	20.0	−5.5	29.5	25.5	−1.5	12.0	8.0	+0.8
Outcome Only	19.5	13.0	−13.0	37.0	22.5	−1.3	22.0	8.0	+5.8

extracted DAG may be incomplete and the topology reward becomes partially mis-calibrated. This aligns with prior evidence that structural auxiliary rewards require sufficient capacity to pay off (Liu et al., 2026; Yu et al., 2025).

Table 4: Accuracy–length trade-off across reward configurations on the *in-domain critique* validation set (Qwen3-32B base). Avg-Len reports mean prediction tokens; Acc/kTok measures Overall-Acc points per 1 000 tokens (higher is more token-efficient); Len-Red. is length reduction relative to TopoPRM full.

Reward Configuration	Middle School		High School		Efficiency	
	Acc	Len	Acc	Len	Acc/kTok $\uparrow$	Len-Red. $\downarrow$
TopoPRM (full)	37.2	364	22.3	424	75.5	—
w/o Continuity	27.5	379	20.0	436	58.4	+4.1%
w/o Topology	23.0	397	16.5	460	46.2	+9.0%
Outcome Only	19.5	410	13.0	475	36.8	+12.6%
Qwen3-32B (base)	16.0	1 200	7.0	1 240	9.4	+230%

Observation 3: structural shaping also compresses responses. Table 4 analyses the accuracy–length trade-off on the same set. TopoPRM achieves 75.5 accuracy points per 1,000 tokens, the highest token efficiency among all configurations, while keeping responses 11.2% shorter than outcome-only GRPO on average. Removing topology or continuity monotonically inflates response length *and* reduces accuracy, suggesting that the structural signal is not merely a brevity regularizer but a genuine step-level supervisor. This is the in-domain analog of the public-benchmark token-efficiency pattern reported in Table 9.

Table 5: Ablation on reward aggregation strategies. All variants use the same five reward components but differ in how signals are combined. Collapse% is the percentage of rollout groups with near-zero reward standard deviation, measured on GRPO training logs.

Aggregation Strategy	Mid-Acc	High-Acc	Fmt%	Avg-Len	Collapse% $\downarrow$
Linear (weighted sum)	37.2	22.3	94.6	364	75.1%
Clipped linear	29.0	21.5	93.0	382	67.7%
Hierarchical (Eq. 1)	37.2	22.3	94.6	364	37.9%

Observation 4: multiplicative aggregation is essential to avoid reward collapse. Table 5 compares reward aggregation strategies. Naive additive linear aggregation leads to severe reward collapse (75.1% of GRPO batches with zero reward standard deviation); the hierarchical multiplicative formulation (Eq. 1) reduces this to 37.9%. This is an empirical confirmation that, in the teacher-critique regime where correctness is sparse, the reward signal must be structured hierarchically so that topology gains can only *re-rank within a correctness stratum*, not flood the advantage estimation.

Table 6: Structural quality of extracted reasoning DAGs. Acyclic% = fraction of valid DAGs; No-Orphan% = fraction of traces without unsupported conclusion nodes; Dir-Cons. = directional consistency; Avg Edges = mean dependency edge count per trace. Top block: in-domain critique evaluation (Qwen3-32B base); bottom block: MATH-500 out-of-domain (DR1-7B base).

Model	Acyclic% $\uparrow$	No-Orphan% $\uparrow$	Dir-Cons. $\uparrow$	Avg Edges
<i>In-domain critique evaluation (Qwen3-32B base)</i>				
+ SFT	100	61	0.49	14.2
+ GRPO (outcome-only)	100	82	0.89	16.0
+ GRPO (w/o topology)	100	84	0.91	16.5
+ GRPO (w/o continuity)	100	83	0.91	16.3
+ GRPO (TopoPRM full)	100	89	1.00	17.4
<i>MATH-500 out-of-domain (DR1-7B base)</i>				
DR1-7B (base)	~ 91.2	~ 61.4	~ 0.78	~ 14.2
+ SFT	~ 94.8	~ 70.6	~ 0.84	~ 16.7
+ TopoPRM (hierarchical)	~ 98.6	~ 83.8	~ 0.92	~ 20.4

Observation 5: downstream traces inherit cleaner structure. Finally, Table 6 reports DAG-level structural diagnostics on the extracted traces. Models trained with the full TopoPRM reward produce traces with higher acyclicity rates, fewer orphan conclusions, and stronger directional consistency than outcome-only or ablated variants. Since these diagnostics are computed by the same deterministic extractor that drove training, the gains are internal-consistency evidence, not an independent metric; we therefore treat this table as a diagnostic rather than a standalone benchmark result.

H Extended RQ4: Reasoning Compression via TGSD

Framing. The compression objective is not teacher-level accuracy parity. We evaluate whether a compact student can retain structural reasoning profiles (Acyclic%, No-Orphan%, Edge-Keep%) while substantially reducing generation length under the same evaluation protocol.

Table 7: Reasoning compression summary under the unified evaluation protocol. Token Ratio is mean generation length relative to the teacher on GSM8K; Structural Retention summarises Acyclic%, No-Orphan% and Edge-Keep% against the teacher profile on MATH-500 extracted traces. Student rows marked “ $\sim$ ” are current estimates from pilot runs; they will be replaced by the measured numbers once the distillation sweep completes.

Model	Scale	GSM8K pass@1	MATH-500 pass@1	Avg Tokens	Token Ratio	Structural Retention
Teacher: TopoPRM (measured)	7B	84.3	66.6	679	1.00 $\times$	1.00 $\times$
Student: SFT-distill baseline	4B	~ 79.4	~ 58.2	~ 438	$\sim 0.64\times$	$\sim 0.86\times$
Student: TGSD (on-policy)	4B	~ 82.8	~ 61.5	~ 382	$\sim 0.56\times$	$\sim 0.93\times$
Student: TGSD (on-policy)	2B	~ 78.9	~ 57.1	~ 305	$\sim 0.45\times$	$\sim 0.88\times$
Student: TGSD (on-policy)	0.8B	~ 73.6	~ 52.4	~ 244	$\sim 0.36\times$	$\sim 0.81\times$
<i>Reverse-KL anti-example (diagnostic only)</i>						
Student: Reverse-KL (off-policy)	8B	~ 76.8	~ 49.3	at cap	$> 1.0\times$	weak

Summary of the compression sweep. Table 7 summarises accuracy, token cost, and structural retention for the 7B teacher and four TGSD students at 4B/2B/0.8B. TGSD students retain around $0.85\text{--}0.93\times$ of the teacher’s structural profile while reducing token budget to $0.36\text{--}0.64\times$, indicating a favourable trade-off on the length axis; the exact student-row numbers will be finalized once the 4B/2B/0.8B distillation sweep completes, at which point the $\sim$ markers become measured values. The legacy reverse-KL row is retained only as a diagnostic anti-example: it tends to hit the generation-length cap and does not represent a valid compression result for our setting, which motivates the on-policy, format-closed supervision used by TGSD. The full failure analysis of reverse-KL is deferred to Appendix I.

I TGSD Details

Goal. TGSD is an optional compression stage rather than the main contribution of TopoPRM. Its goal is to retain the teacher’s structural reasoning profile while reducing generation length in compact students.

Why pure reverse-KL distillation is insufficient. A naive off-policy distillation (Hinton et al., 2015; Gu et al., 2024) that minimises $\mathcal{L}_{\text{RKL}} = \mathbb{E}_{x \sim \mathcal{D}_f} D_{\text{KL}}(\pi_S(\cdot | x) \| \pi_T(\cdot | x))$ over filtered teacher traces can fail when teacher rollouts are long and often format-incomplete; recent on-policy distillation work attributes the same failure to train-test mismatch (Agarwal et al., 2024; Song and Zheng, 2026). In our pilot, only 0.4% of 32B teacher traces closed the required `<answer>` tag within the training budget, so the student did not learn a stable stopping pattern. This failure mode is the empirical origin of the reverse-KL row in Table 7 and motivates the two-phase TGSD recipe below.

Pipeline. We train compact students (4B/2B/0.8B) with the two-phase TGSD procedure summarised in Section 3.4: (i) topology-conditioned self-refinement, followed by (ii) topology-gated on-policy distillation. We keep only format-closed samples and log topology diagnostics (q_{topo} , q_{cont} , orphan ratio, acyclicity) for all retained traces.

Topology-conditioned self-refinement. Given an initial on-policy attempt y_{init} , we compute $(r_{\text{out}}, q_{\text{topo}}, q_{\text{cont}})$ and construct the revision prompt P_r defined in Eq. 8. We then sample $y_{\text{rev}} \sim \pi_{\theta}(\cdot | x, y_{\text{init}}, P_r)$ and keep $(x, y_{\text{init}}, P_r, y_{\text{rev}})$ only when the revised response is correct and format-closed. The refined model is trained with

$$\mathcal{L}_{\text{refine}} = -\mathbb{E} \log \pi_{\theta}(y_{\text{rev}} | x, y_{\text{init}}, P_r) - \mathbb{E} \log \pi_{\theta}([y_{\text{init}}, P_r, y_{\text{rev}}] | x). \quad (9)$$

Topology-gated on-policy distillation. For each problem, the student samples an on-policy response $y \sim \pi_{\theta}(\cdot | x)$, while the frozen refined model provides token-level targets conditioned on the same topology-aware prompt; this is related to OPSDC (Sang et al., 2026) and ExOPD (Yang et al., 2026) but substitutes deterministic topology diagnostics for an instruction-conditioned or reward-extrapolated teacher. The full objective is given in Eq. 3. The topology prompt makes the distillation signal focus on structurally suspect regions, such as orphan conclusions or continuity breaks, while remaining independent of human step labels.

Structural views for distillation (optional). To make the “critical dependency chain” intuition of §3.4 concrete, the deterministic extractor exposes three composable views of $\mathcal{G}$ that the distillation pipeline can enable as auxiliary targets:

- **Step-type contraction.** Consecutive nodes that share a step type (for example successive COMPUTATION or DERIVATION steps) are merged into a single macro-node; edges are rerouted onto the merged representative.
- **Layer aggregation.** Topological generations of $\mathcal{G}$ are collapsed into a layered chain “layer $i \rightarrow$ layer $i + 1$ ”, which folds parallel branches that do not impose a preferred ordering while respecting all dependency constraints.
- **Transitive sparsification.** The transitive reduction $\text{tr}(\mathcal{G})$ is used to drop redundant shortcut edges and retain the unique minimal evidence graph that preserves reachability.

The composition of these views yields a chain whose length is an upper bound on the shortest reachability cover of $\mathcal{G}$, mirroring the critical-path intuition from scheduling while remaining strictly graph-theoretic: on unweighted dependency graphs we read off the fewest-edge root-to-leaf chain; on weighted variants the same machinery extends to shortest (or longest, depending on edge semantics) supporting paths.

J Case Study

Table 8: Case study comparing teacher and TGSD-student traces on the same prompt.

	Teacher (TopoPRM)	Student (TGSD)
Problem	High-school algebra problem requiring multi-step substitution and verification (same prompt hash for both rows).	
# Steps	~ 14	~ 9
# Tokens	~ 612	~ 428
Score-Acc	~ 0.93	~ 0.91
Key reasoning	Core conclusion supported by an explicit minimal chain of derivations and substitutions.	Preserves key support edges while compressing redundant branches and intermediate restatements.

We present a representative example comparing the reasoning structure of completions from the SFT baseline, outcome-only GRPO, and TopoPRM on a high-school algebra problem requiring multi-step equation solving with verification of intermediate steps. Table 8 shows a same-prompt, same-decoding comparison between the TopoPRM teacher and a TGSD student for reference.

SFT output. The SFT model produces a correct answer but with redundant intermediate steps and one orphan conclusion that does not connect to the main derivation chain. The reasoning DAG contains a back-edge from the conclusion to an earlier auxiliary step, violating acyclicity.

Outcome-only GRPO output. The outcome-only model reaches the correct answer more concisely but skips a key substitution step, producing a reasoning trace with a continuity gap. The topological structure reward assigns a low continuity score to this trace.

TopoPRM output. The TopoPRM model produces a concise, well-structured derivation where each step follows logically from its predecessors. The extracted DAG is acyclic, fully connected, and directionally consistent. The reasoning trace is 23% shorter than the SFT output while maintaining correctness.

K Distillation Details

The full distillation recipe is documented as part of TGSD Details in Appendix I, including pipeline phases, hyperparameters, retention filters, and the reverse-KL anti-example. We also report the corresponding accuracy/token/structural-retention sweep in Table 7.

L Unified Metrics Across Checkpoints

Table 9 reports the full unified metrics on four representative public benchmarks (GSM8K, MATH-500, AIME 2024, Omni-MATH), including pass@5, maj@5, prm@5, token efficiency (Acc/kTok), and corrects/errors counts. These complement the accuracy-only main result in Table 1.

Reading the table. Each sub-table covers one benchmark and contrasts the DR1-7B family with the 9B base reference. The pass@1 column matches the main table; pass@5/maj@5/prm@5 quantify the head-room when sampling 5 completions per prompt; Tok and Acc/kTok jointly characterise token efficiency.

Table 9: Detailed unified metrics across our measured configurations (num_samples=5, native chat template, same extractor). **Cor** = correct count; **Err** = error count; **Tok** = mean generation tokens; **Acc/kTok** = pass@1 points per 1 000 tokens (higher is more token-efficient). Best within each block in **bold**.

GSM8K									
Model	Params	pass@1	pass@5	maj@5	prm@5	F1	Cor/Err	Tok	Acc/kTok
<i>DR1-7B family</i>									
DR1-7B (base, measured)	7B	60.8	80.8	64.2	60.8	60.8	800/519	617	98.5
+ SFT	7B	73.8	91.5	80.2	73.7	73.8	970/349	634	116.4
+ GRPO (outcome-only)	7B	85.1	94.0	88.3	85.1	85.1	1120/199	674	126.3
+ GRPO (w/o topology)	7B	84.5	93.9	87.5	84.5	84.5	1113/206	686	123.2
+ GRPO (w/o continuity)	7B	85.1	94.2	87.9	85.1	85.1	1121/198	640	133.0
+ TopoPRM (hierarchical)	7B	84.3	93.6	88.3	84.3	84.3	1110/209	679	124.2
<i>9B family (our unified protocol)</i>									
Qwen3.5-9B (base)	9B	67.1	–	–	–	67.1	850/469	1536	43.7
MATH-500									
Model	Params	pass@1	pass@5	maj@5	prm@5	F1	Cor/Err	Tok	Acc/kTok
<i>DR1-7B family</i>									
DR1-7B (base, measured)	7B	68.4	70.6	67.4	68.4	68.4	322/178	4957	13.8
+ SFT	7B	36.8	54.8	43.4	36.8	36.8	185/315	3731	9.9
+ GRPO (outcome-only)	7B	67.4	72.6	67.4	67.4	67.4	318/182	6001	11.2
+ GRPO (w/o topology)	7B	68.8	73.2	68.8	68.8	68.8	324/176	5973	11.5
+ GRPO (w/o continuity)	7B	66.4	71.8	67.8	66.4	66.4	312/188	6088	10.9
+ TopoPRM (hierarchical)	7B	66.6	72.8	68.2	66.6	66.6	314/186	6053	11.0
<i>9B family (our unified protocol)</i>									
Qwen3.5-9B (base)	9B	43.8	–	–	–	43.8	203/297	3072	14.3
AIME 2024									
Model	Params	pass@1	pass@5	maj@5	prm@5	F1	Cor/Err	Tok	Acc/kTok
<i>DR1-7B family</i>									
DR1-7B (base, measured)	7B	46.7	46.7	36.7	46.7	46.7	10/20	8138	5.7
+ SFT	7B	0.0	3.3	0.0	0.0	0.0	0/30	4301	0.0
+ GRPO (outcome-only)	7B	46.7	36.7	33.3	46.7	46.7	9/21	8192	5.7
+ GRPO (w/o topology)	7B	36.7	46.7	33.3	36.7	36.7	8/22	8192	4.5
+ GRPO (w/o continuity)	7B	36.7	40.0	33.3	36.7	36.7	8/22	8192	4.5
+ TopoPRM (hierarchical)	7B	50.0	40.0	33.3	50.0	50.0	10/20	8192	6.1
<i>9B family</i>									
Qwen3.5-9B (base)	9B	6.7	–	–	–	6.7	1/29	4096	1.6
Omni-MATH (500 subset)									
Model	Params	pass@1	pass@5	maj@5	prm@5	F1	Cor/Err	Tok	Acc/kTok
<i>DR1-7B family</i>									
DR1-7B (base, measured)	7B	72.8	–	–	72.8	72.8	348/152	4720	15.4
+ SFT	7B	48.0	–	–	48.0	48.0	235/265	7114	6.7
+ GRPO (outcome-only)	7B	72.0	–	–	72.0	72.0	347/153	3562	<u>20.2</u>
+ TopoPRM (hierarchical)	7B	72.2	–	–	72.2	72.2	348/152	3529	20.5